09-10-25

UNIT-3 (CONTINOUS DISTRIBUTION)

Uniform Distribution

Uniform Distribution is a Continous distribution

The random variable 'x' is said to follow uniform distribution if its probability density function f(x) is

$$f(x) = \begin{cases} k, & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

w.k.T $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_a^b k\,dx = 1$$

$$[kx]_a^b = 1$$

$$\Rightarrow k(b-a) = 1$$

$$\boxed{k = \frac{1}{b-a}}$$

$\therefore$ p.d.f is $F(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$

$$\text{mean} = \int_a^b x.f(x)\,dx$$

$$= \int_a^b x.\frac{1}{b-a}\,dx = \frac{1}{b-a}\left(\frac{x^2}{2}\right)_a^b$$

$$= \frac{1}{2(b-a)}(b^2-a^2)$$

$$\boxed{\text{mean} = \frac{b+a}{2}}$$

$$\boxed{\text{variance} = \frac{(a-b)^2}{12}}$$

$$\text{variance} = \int_a^b x^2 f(x)\,dx - \mu^2$$

$$= \int_a^b x^2 \frac{1}{b-a}\,dx - \left(\frac{b+a}{2}\right)^2$$

$$= \frac{1}{b-a}\left(\frac{x^3}{3}\right)_a^b - \frac{(b+a)^2}{4}$$

$$= \frac{1}{3(b-a)}(b^3-a^3) - \frac{(b+a)^2}{4}$$

$$= \frac{(b-a)(b^2+ab+b^2)}{3(b-a)} - \frac{(a^2+b^2+2ab)}{4}$$

$$= \frac{4(b^2+ab+b^2)-3(a^2+b^2+2ab)}{12} = \frac{a^2+b^2-2ab}{12} = \frac{(a-b)^2}{12}$$

Q) Subway trains runs on a certain line every half an hour between midnight and 6 AM in the morning. what is the prob that a man entering a station at random time will have to wait atleast 20 minutes.

let x represent man's waiting time for the next train clearly x is a continous random variable which is uniformly distributed in the interval (0,30) when a man arrives at the station.

The probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a<x<b \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore f(x) = \begin{cases} \frac{1}{30-0} & 0<x<30 \\ 0 & \text{otherwise} \end{cases}$$

P(man to wait for atleast 20 minutes) = $P(x \geq 20)$ $\rightarrow \int_{20}^{30} f(x)\,dx$

$$= \int_{20}^{30} \frac{1}{30}\,dx$$

$$= \frac{1}{30}(30-20)$$

$$\boxed{P(x \geq 20) = 1/3}$$

2Q) If 'x' is uniformly distributed over (-a,a) then find 'a' such that $P(x>1) = 1/3$

we know that prob density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a<x<b \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore f(x) = \begin{cases} \frac{1}{2a}, & -a<x<a \\ 0 & \text{otherwise} \end{cases}$$

$P(x>1) = 1/3$

$$\frac{1}{3} = \int_{1}^{a} f(x)\,dx$$

$$\frac{1}{3} = \int_{1}^{a} \frac{1}{2a}\,dx$$

$\frac{1}{3} = \frac{1}{2a}[a-1]$

$\frac{2a}{3} = a-1$

$1 = a/3$

$\Rightarrow \boxed{a = 3}$

Normal Distribution:

Normal Distribution is a continous distribution. It is also called as Gaussian Distribution. It is very important in statistical theory, because of 2 reasons

1. other distributions like binomial, poisson, Hypergeometric, ... (fail, you can replace with Normal) can be derived from Normal distribution.

2. Even if the variable is not normally distributed, it can be brought to the normal form, by simple transformation in the variable.

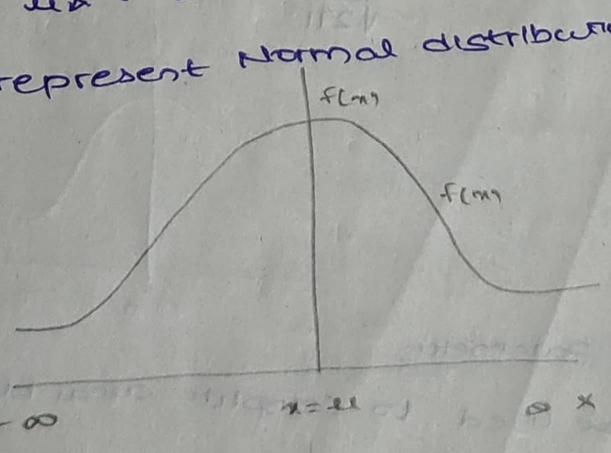
NOTE-1:- Normal distribution acts as the limiting case to Binomial distribution... continuation in next pg

- The r.v 'x' is said to be normally distributed, if it takes all the possible values from $-\infty < x < \infty$ & its prob density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where } -\infty < x < \infty,\ -\infty < \mu < \infty,\ \sigma > 0$$

Characteristics or S.T mean = median = mode

- Here μ & σ are the parameters of Normal distribution we represent Normal distribution as $N(\mu, \sigma)$



Normal Distribution curve

- The graph of the Normal distribution is called a Normal curve.

- It is a bell shaped curve symmetrical about the mean 'μ'.
- From the fig, it is clear that $f(x)$ has maximum at $x = \mu$ therefore mode is μ (from def)
- ~~It is~~ Clear the line $x = \mu$ divides the region above x-axis into 2 equal parts, so therefore the median is μ.
- So, we can say that mean = median = mode for normal distribution.
- The total area under this normal probability curve is 1.

include z-distribution too in characteristics

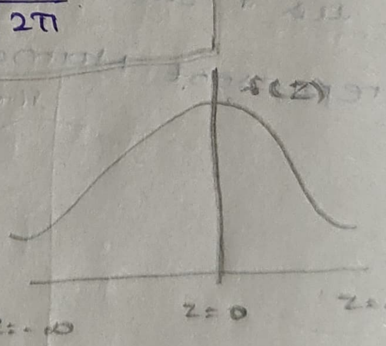
Standard normal form :-

If 'x' is a normally distributed r.v with mean μ and (variance σ^2 or) std deviation σ, then $\boxed{z = \frac{x-\mu}{\sigma}}$ is also a normally distributed r.v with mean = 0, std deviation = 1.

Here, z is called std. normally distributed variable
for std normal distributed variable
and $z = \frac{x-\mu}{\sigma}$ is called std. normal form

The probability density function in standard normal form is given by $\boxed{f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-z^2/2}}$

↳ no need of integration
& could use areas given in table



As f(z) is free from parameters μ and σ, it can be used to compute area by using standard normal distribution table

z-distribution

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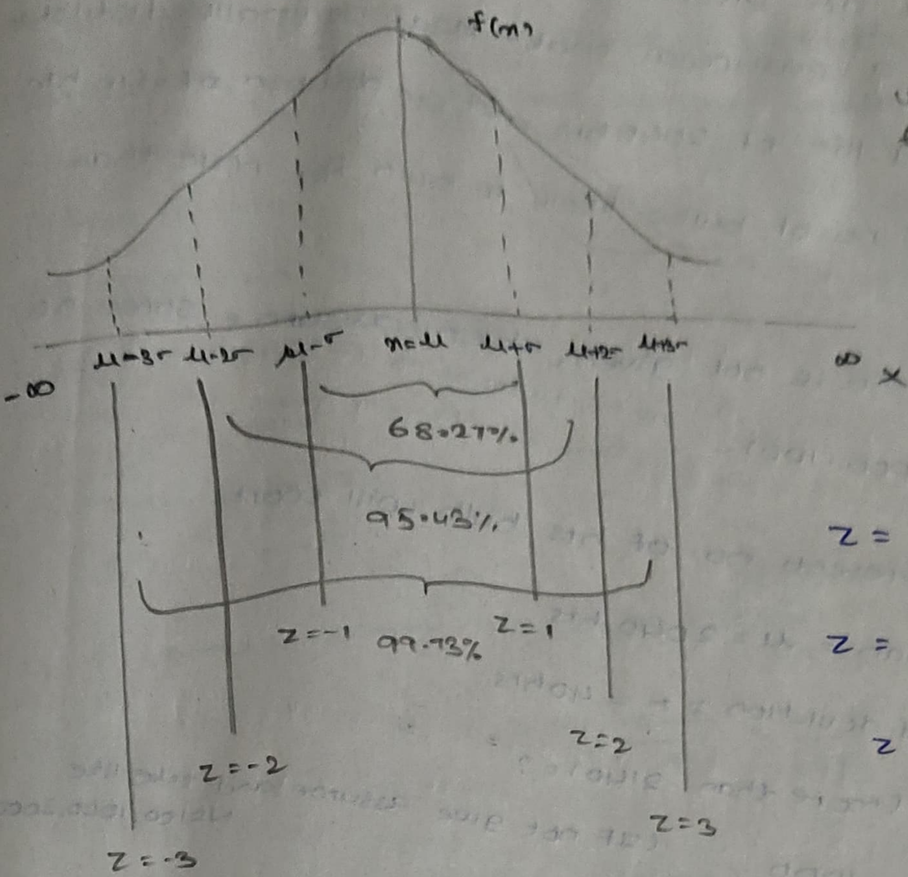


values are taken till 3 & 3+

∴ 100% area is covered & 100% it is symmetric

[illegible]

$$z = \frac{x-\mu}{\sigma}$$

$$z = \frac{\mu-\sigma-\mu}{\sigma}$$

$$z = -1$$

Properties:-

i) $F(z) = P(Z \leq z) = \int_{-\infty}^{z} f(z)\,dz$

ii) $F(-z) = 1 - F(z)$

iii) $\int_{-\infty}^{\infty} f(z)\,dz = 1$

iv) $\int_{-\infty}^{0} f(z)\,dz = \int_{0}^{\infty} f(z)\,dz = 0.5$

v) $\int_{-a}^{a} f(z)\,dz = 2\int_{0}^{a} f(z)\,dz$

vi) $\int_{-a}^{0} f(z)\,dz = \int_{0}^{a} f(z)\,dz$

NOTE-1:- Normal distribution acts as the limiting case to binomial distribution when $n \to \infty$ and $p \approx 1/2$

NOTE-2:- Normal distribution acts as the limiting case to poisson when $\lambda \to \infty$

Q) In a test on electrical bulbs, it was found that the life of a particular make was normally distributed with an avg life of 2040 hrs and std deviation of 40 hrs. Estimate the no. of bulbs likely to burn for more than 2140 hrs.

(If such no is not given, then assume some no such as 1000, 100)..

we need a total no to calculate the Prob so when not given, we need to assume

let 'x' represents no. of hrs bulb will work

Given mean $= \mu = 2040$ hrs

std deviation $= \sigma = 40$ hrs

P(more than 2140) = ?

(If not give assume and take like N=100, 1000, 200)

let N = 1000

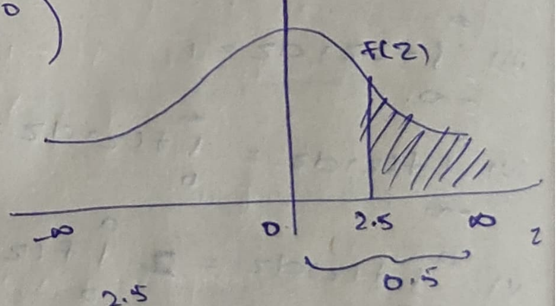
$= P(x > 2140)$

standard normal form $z = \frac{x-\mu}{\sigma}$

$=$

$= P\left(\frac{x-\mu}{\sigma} > \frac{2140-\mu}{\sigma}\right)$

$= P\left(z > \frac{2140-2040}{40}\right)$



$= P(z > 2.5)$

$= \int_{2.5}^{\infty} f(z)\,dz$

$= \int_{0}^{\infty} f(z)\,dz - \int_{0}^{2.5} f(z)\,dz$

$= 0.5 - 0.4938$

$= 0.0062$

2.5 → 2.5 under 0
2.52 → 2.5 under 2

No. of bulbs working for more than 2140 hrs is

$N \times 0.0062 = 1000 \times 0.0062$

$= 6.2$

20.) The weekly wages of 1000 workers are normally distributed with a mean of ₹70 & std deviation of ₹5. Estimate the no. of workers whose weekly wages will be

i) b/w 70 & 72

ii) b/w 69 & 72

iii) more than 80.

let X - weekly wages

$N = 1000$

mean $= \mu = 70$

S.D $= \sigma = 5$.

i) P (weekly wages lying b/w 70 & 72) $= P(70 < X < 72)$ standardise them to use normal form.

$$= P\left(\frac{70-\mu}{\sigma} < \underbrace{\frac{X-\mu}{\sigma}}_{Z} < \frac{72-\mu}{\sigma}\right)$$

$$= P\left(\frac{70-70}{5} < Z < \frac{72-70}{5}\right)$$

$$= P(0 < Z < 0.4)$$

$$= \int_0^{0.4} f(z)\,dz$$

$$= 0.1554$$

No. of weekly wages lying b/w 70 & 72 is $N \times 0.1554$

$= 1000 \times 0.1554$

$= 155.4 \approx 155$

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ii) $P(69 < X < 72)$

$$= P\left(\frac{69-\mu}{\sigma} < \underbrace{\frac{X-\mu}{\sigma}}_{Z} < \frac{72-\mu}{\sigma}\right)$$

$$= P(-0.2 < Z < 0.4)$$

$$= \int_{-0.2}^{0} f(z)\,dz + \int_0^{0.4} f(z)\,dz = 0.1554 = 0.2347$$

$$- \int_0^{0.2} f(z)\,dz + 0.0793 +$$

No. of weekly wages lying b/w 69 & 72 is $N \times 0.2347$

$= 1000 \times 0.2347$

$= 234.7 \approx 235$

iii) $P(X > 80)$

$= P\left(\frac{x-\mu}{\sigma} > \frac{80-70}{5}\right)$

$= P(Z > 2)$

$= 0.5 - P(Z < 2)$

$= 0.5 - \int_0^2 f(z)\,dz$

$= 0.5 - 0.4772$

$= 0.0228$

no. of persons whose weekly wages are > 80.

$\Rightarrow 28$

$\Rightarrow \cancel{1000 \times 4.28\%} = \cancel{42.28}$

10m PYQ

1. A sales tax officer has reported that avg sales of 500 business that he has to deal with during a year is ₹ 36000. with a std deviation 10000 assuming that the sales in the business are normally distributed.

Find the i) no. of business as the sales of which are 40000

ii) the % of business sales which are likely to range b/w 30000 & 40000.

let x represent sales

Given that $\mu = 36000$

SD, $\sigma = 10000$

$N = 500$

i) P(sales of 40000) $= P(X \geq 40{,}000)$

$= P\left(\frac{x-\mu}{\sigma} \geq \frac{40000-36000}{10000}\right)$

$= P\left(\frac{x-\mu}{\sigma} \geq 0.4\right)$

$= P(Z \geq 0.4)$

$= 0.5 - P(Z \leq 0.4)$

$= 0.5 - \int_0^{0.4} f(z)\,dz$

$= 0.5 - 0.1554$

$= 0.3446$

no. of sales > 40000 is 0.3446×500

$\simeq 172.3 \approx 172$

ii) p(Sales lying b/w 30000 & 40000) = $P(30000 < x < 40000)$

$$= P\left(\frac{30000-36000}{10000} < \frac{x-\mu}{\sigma} < \frac{40000-36000}{10000}\right)$$

$$= P(-0.6 < z < 0.4)$$

$$= \int_{-0.6}^{0} f(z)dz + \int_{0}^{0.4} f(z)dz$$

$$= \int_{0}^{0.6} f(z)dz + \int_{0}^{0.4} f(z)dz = 0.2257 + 0.1554$$

$$= 2\int_{0}^{0.4} f(z)dz + \int_{0.4}^{0.6} f(z)dz$$

$$= 2[0.1554] + [0.2257 - 0.1554]$$

$$= 0.3811$$

$$= 38\%$$

2 The marks obtained in maths by 1000 students is normally distributed with mean 78% and std deviation 11%. Determine i) How many students got marks above 90%

ii) what was the heighest mark obtained by the lowest 10%

iii) within what limits did the middle of 90% of students lie.

let 'x' represent % of marks

Given mean = $\mu = 78\%$

Std deviation = $\sigma = 11$ $N = 1000$

i) $P(x > 90)$

$$P\left(\frac{x-\mu}{\sigma} > \frac{90-78}{11}\right)$$

$$P\left(z > \frac{12}{11}\right) = P(z > 1.09)$$

$$= 0.5 - P(z < 1.09)$$

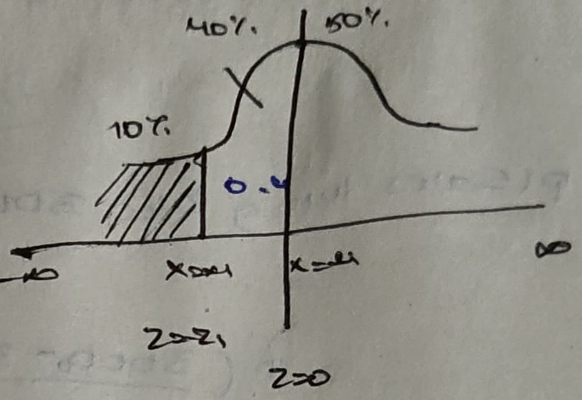
$$= 0.5 - \int_{0}^{1.09} f(z)\,dz = 0.5 - 0.3621$$

$$= 0.1379$$

$\Rightarrow 0.1379 \times 1000 = 137.9$

≈ 138

iii) from fig



$P(-z_1 < z < 0) = 0.4$

$\int_{-z_1}^{0} f(z)\,dz = 0.4$

$-z_1 = \frac{x_1 - \mu}{\sigma}$ $\quad -z_1 = \frac{x_1 - 78}{11}$

$\int_{0}^{z_1} f(z)\,dz = 0.4$

$z_1 = 1.285$ consider the one at less dist
both at same range ⇒ take avg

$-z_1 = \frac{x_1 - \mu}{\sigma}$

$-z_1\sigma = x_1 - \mu$

$\mu - z_1\sigma = x_1$

$78 - (1.28)(11) = x_1$

$x_1 = 63.92 \simeq 64$